Table 2: Performance (MAE) comparison with the baselines on band gap prediction.

Model	#Parameters	Band gap (eV) Validation set ↓	Test set ↓
Structure-based models			
CGCNN (Xie and Grossman, 2018)	-	0.301	0.293
MEGNet (Chen et al., 2019)	-	0.300	0.304
ALIGNN (Choudhary and DeCost, 2021)	-	0.249	0.250
Text-based models			
MatBERT (zero-shot)	109.5M	1.325	1.048
MatBERT		0.244	0.249
LLM-Prop (zero-shot-512 tokens)	37M	1.022	1.070
LLM-Prop (512 tokens)		0.238	0.249
LLM-Prop (zero-shot)		1.117	1.031
LLM-Prop		**0.229**	**0.241**

Table 3: Performance (MAE) comparison with the baselines on volume prediction.

Model	#Parameters	Volume ($Å^3$/cell) Validation set ↓	Test set ↓
Structure-based models			
CGCNN	-	188.834	188.368
MEGNet	-	297.948	303.187
ALIGNN	-	129.580	126.486
Text-based models			
MatBERT (zero-shot)	109.5M	483.089	482.578
MatBERT		49.761	53.282
LLM-Prop (zero-shot-512 tokens)	37M	483.009	485.378
LLM-Prop (512 tokens)		49.063	53.880
LLM-Prop (zero-shot)		482.863	485.396
LLM-Prop		**42.259**	**44.553**

predicting whether the band gap is direct or indirect, LLM-Prop also outperforms the best-performing baseline (ALIGNN) on the validation set with an approximate 5% improvement and improves upon the best-performing baseline on the test set (CGCNN) by 3% (see Table 5). Interestingly, finetuning LLM-Prop with just 512 tokens as an input sequence length also outperforms all GNN-based baselines, despite not being able to account for enough context from the text descriptions (see Table 2, 3, and 5). Unfortunately, LLM-Prop is not able to get better performance when we do zero-shot predictions since T5 was not explicitly trained on in-domain data. However, the finetuned LLM-Prop (LLM-Prop-transfer) on band gap exhibits strong transfer learning ability compared to other models when transferred to volume prediction and vice-versa (see Table 4).

LLM-Prop vs MatBERT. For fair comparison, we first consider LLM-Prop and MatBERT when they are trained on the same sequence length (512 tokens). LLM-Prop